

AACE International Recommended Practice No. 48R-06

SCHEDULE CONSTRUCTABILITY REVIEW

TCM Framework: 7.2 – Schedule Planning and Development
11.5 – Value Management and Value Improving Practices (VIPs)

Acknowledgments:

Edward E. Douglas, III CCC PSP (Author)
Ricardo Accioly
Rodney B. Adams, CCE
Zia Akhtar
Kenneth Baker
Michael P. Bomi
Timothy T. Calvey, PE PSP
Paul L. Conant, Jr.
Douglas A. Findley, CCC PSP

Ricardo Garcia da Roza
John K. Hollmann, PE CCE CEP
John J. MacDonald
Robert C. Powell, CCE
Benjamin Price
Mahmoud M. Saleh
Hannah E. Schumacher, PSP
Donald F. Sulzer
Ronald M. Winter, PSP

SCHEDULE CONSTRUCTABILITY REVIEW

TCM Framework: 7.2 – Schedule Planning and Development, 11.5 – Value Management and Value Improving Practices (VIPs)



August 28, 2009

INTRODUCTION

Purpose

This recommended practice (RP) is intended to serve as a guideline, not establish a standard for schedule constructability reviews. This recommended practice describes the schedule constructability review (SCR) process and some of the recommended planning that should be considered when developing a construction project execution-phase schedule. This recommended practice includes a suggested review process for the construction project schedule. This RP was written as a stand alone document however it can be used as a companion guideline with the AACE Recommended Practice 30R-03 *Implementing Project Constructability*.^[26]

RECOMMENDED PRACTICE

Construction contract schedule specifications frequently require reviews of the project baseline schedule with very little guidance about how that review process should be accomplished. There may be contract language that suggests the “schedule should be consistent with the project scope” and that there should be an appropriate level of detail to facilitate integration of the various contractors or trades^[21]. Numerous technical articles have been published that recommend analyzing the project cost and schedule. Several of those articles describe a recommended process for the detailed analysis of construction project cost estimates. It was difficult to locate comparable information from these technical resources that describes the process to analyze the construction project schedule. Constructability review checklists that were located assume an understanding of what is required by an SCR and barely mention the need to perform a construction schedule assessment. A schedule constructability review is performed to analyze and assess the feasibility of the construction plan. The SCR compares the planned sequence of work with the project scope as defined in the work breakdown structure (WBS), site requirements, and the specific needs of the client. This recommended practice will focus on constructability concepts influencing the construction execution planning and a suggested review assessment for the construction project schedule.

Background

The Construction Industry Institute (CII) and AACE International define “constructability [as] the optimum use of construction knowledge and experience in planning, design/engineering, procurement, and field operations to achieve overall project objectives.”^[4, 25]

A constructability review is a structured review of the plans and specifications with the focus on the buildability, biddability and efficiency of construction. Constructability reviews are performed to assure consistency between design, fabrication, and installation. These reviews identify errors, conflicts, and omissions and as a result of constructability reviews future costly field changes can be minimized.

The application of constructability concepts to capital construction projects have been reported to provide a return on investment of at least ten to one (10:1). Implementing project constructability concepts reduces project costs and schedule while having a positive impact on quality and safety.^[4]

A constructability review is the process of evaluating the construction documents (design drawings, and technical specifications) for clarity, consistency, completeness, and ease of construction to achieve overall project objectives. The objective of this review is to provide clarity, consistency, and completeness of the contract documents to facilitate construction bidding, administration, and interpretation to achieve overall project objectives.^[6]

Several factors influence the effectiveness of a constructability review program: the project delivery system, project contracting strategy, procurement strategy for material and equipment, as well as the completeness of the project scope definition. Constructability reviews generally consist of:

- construction involvement during project design;
- detailed project scope review;
- detailed review of construction plans and specifications;
- execution plan development and review;
- detailed schedule and budget review; and
- development of identified alternatives.

Constructability reviews are performed as a cross check of construction documents for accuracy, completeness, and systems design coordination issues. "During the constructability review, the focus is on improving:

- consistency, clarity and completeness of the construction drawings and specifications
- consistency, applicability, enforceability, and comprehensiveness of the general condition ("front end" documents)
- applicability of construction installation technology, methodology or materials
- consistency between plans and site conditions
- identification of project-specific issues, their probable consequences, and proposed mitigation recommendation"^[23]

According to studies the implementation of constructability reviews and assessments at all stages of the project from conception to commissioning have been shown to provide opportunities to identify both cost and schedule savings for the project. Early involvement of construction industry experience can reduce or eliminate problems, by identifying potential conflicts and facilitate a balance between production requirements and building site constraints.^[18]

While the implementation of a constructability review program can be difficult it can be even more difficult to establish a formal schedule constructability assessment program.

Schedule Constructability Review

A project construction schedule should be a comprehensive and realistic plan that represents the specific activities, reasonable durations for the activities, and the planned sequence of work for the project. The logic or sequence of work activities should represent how the project will be built and how the various activities are interrelated. The primary objective of an SCR is to determine if the project schedule is "accurate, logical and achievable."^[1]

A schedule constructability review is intended to assess whether the construction schedule is comprehensive and complete. The focus of an SCR is to assess the following:

- consistency
- clarity
- completeness and reasonableness of the work sequence
- coordination of the schedule with the various engineering disciplines
- coordination of the schedule with the requirements for efficient start-up, and commissioning
- adequacy of lead time for material and equipment procurement
- site restrictions and adequacy of site access

The SCR is intended to disclose problems in the following areas:

- reasonableness of work sequence
- comprehensiveness and completion of construction planning
- coordination and interface among the various craft trades and engineering disciplines
- adequacy of lead time for material and equipment procurement
- site work restrictions and adequacy of site access

Schedule Constructability Reviews During the Various Phases of a Project

The expectations for accuracy, completeness, and the level of construction schedule detail will increase with development of the project through the various phases from the pre-construction planning time line through to the detailed execution schedule for construction. It has been recommended that the SCR process should be implemented in “vertical slices” for each discipline or major feature of work. For example a vertical slice through the site civil work would include all engineering and design disciplines as well as the civil and underground (mechanical & electrical) contractors. Changes and additions would then be implemented before reviewing the next vertical slice. The final review would encompass the entire project with all project participants, contractors, engineering design, and procurement represented. The conceptual or feasibility phase schedule could be expected to consist of a dozen key milestones and possibly a bar chart of the significant project activities. Later into the project pre-construction planning before the execution phase, there would be a high level of detail to identify the hundreds (or possibly thousands) of activities required to complete the construction and installation of the multiple features and systems for the startup and operation of that type of facility. In some industries such as the process industry, there would be a numeric classification assigned to the schedule as the project scope definition is developed.

Design Phase

As project planning and design progresses beyond the conceptual phase, schedule activities will be added to identify the important features of work: site areas to be modified (cleared, excavated, shored, underground utilities, etc.); then key equipment to be installed and major systems to be operationally tested will be included as the schedule is developed during the design phase. Although the design phase construction schedule is intended to ultimately guide the construction contractor (or entity) who will perform the construction work, these schedules have a tendency to represent the work at a higher level (with less detail) and may not portray accurate estimated durations for construction success. The duration inaccuracies in the design phase may occur due to a lack of accurate quantity details. Design phase construction schedules will typically identify key construction milestones and construction phase activities that portray the overall plan for the client. However the phases within these schedules may overlap but generally do not link the specific detailed craft interfaces. Key equipment installations may be included at various construction points, but often the deliveries are inaccurately forecasted due to a failure to verify realistic equipment fabrication and delivery schedules.

The level of detail for a construction schedule can be expected to increase in complexity during the design phase for a construction project. An SCR can be performed at any of the key design completion phases: schematic design, detailed design, etc., corresponding with various stages of construction document completion (30%; 60%; 90% 100%). Traditionally a constructability review will be conducted after the construction documents are complete and prior to contractor bidding. Under an integrated project delivery (IPD) contract, the preconstruction phases are identified differently as follows: conceptualization, criteria design, detailed design, implementation documents, and buy-out phases; however the intent of the SCR at this phase has the same focus the identity of potential coordination issues, availability of specified materials, equipment and commodities, out of sequence work, missed work details, unrealistic activity durations, potential time delays, and inter-contractor coordination points prior to issuing project bid documents.

Procurement Phase

Depending upon the requirements outlined in the request for proposal (RFP), potential construction bidders may be required to provide a detailed procurement/fabrication and construction schedule with their proposal for the construction project. That proposal construction schedule must consider the vendors input as well as the project and site requirements, and the specific needs of the client. An SCR of these potential bidders along with the proposed procurements in relation to their construction schedules can be appropriate and beneficial prior to the construction contract award.

Execution Phase

The project construction execution phase schedule represents the construction entity's means and methods for successfully accomplishing the construction work. A constructability review of the execution phase schedule is intended to compare the schedule with the entire scope of the project construction efforts and should be conducted prior to the owner's acceptance/approval of that construction schedule. The execution phase SCR can assess a variety of items that could affect the time phased plan for performance of the work:

- physical limitations of the work site
- material and equipment "lay down" and storage area requirements
- availability of job site utilities
- interfaces with the public domain, other construction contractors, and other entities
- availability and completeness of design documents and drawings
- limitations and restrictions on hours of operation
- seasonal influences on the work; local holidays and other factors affecting productivity rates
- physical limitations of construction equipment
- physical limitations and "handle-ability" of construction components
- the owner's operational requirements
- interfaces and coordination with utility companies
- availability of labor, equipment and materials; long lead time equipment items
- site safety and security; safety and security of the general public
- traffic impacts
- aesthetics and safety aspects of barriers and buffers
- noise mitigation
- job site cleanliness and maintenance requirements
- construction cost and schedule
- pay item structures and quantities
- quality control, quality assurance, and system testing requirements^[14]

Schedule Constructability Review Considerations

The schedule review must be conducted by someone who is knowledgeable and understands the construction requirements of the project. The schedule review must be conducted by a person or persons with experience in the unique requirements for that type of project whether it is general commercial facility; high rise building with unique vertical construction requirements; horizontal or linear construction such as highways; or specialized projects such as tunnels, bridges, waste water treatment facilities, brownfield sites, sports stadiums, casinos, expensive custom homes, railroads & transit projects, or power plants, etc.

The constructability team reviewers should have experience with similar types of projects to better perform a constructability assessment, to provide a credible evaluation of the sequence of work and the activity durations established for the work. Schedule activity durations are partially established by consideration of the quantities to be installed from the cost estimator's drawing take off, as well as other

project factors that might affect productivity such as site congestion or seasonal impacts which can better be determined by an experienced constructor.

The constructability schedule reviewer is often not a scheduling professional but must have an understanding of good scheduling practices and generally accepted principles for “building” the schedule that accurately represents the sequence and logic of how the project will be built. Some of the important considerations for the constructability schedule reviewer include:

- *Completeness.* The schedule must be comprehensive in scope including all predecessor and successor work activity relationships from the construction notice to proceed (NTP), mobilization, execution and on through demobilization.
- *Coordination.* The schedule must take into account the coordination and interfaces with all project participants.
- *Start-up and commissioning.* The schedule must be integrated with the requirements for start-up, commissioning and transition to efficient operations of the facility and support systems.
- *Identification of work done by others.* Craft trades; subcontractors; vendors and suppliers; and work performed “by others” must be clearly indicated for appropriate clarity in the plan.
- *Control points.* Engineering & design completion requirements should be indicated in the scheduled sequence as a milestone such as “issued for construction (IFC) design documents”.
- *Safety, security and environmental requirements.* The schedule should take into account the special safety, security or environmental requirements above and beyond the generally accepted standards.
- *Long lead items.* Material and equipment procurements should be a major focus during the construction schedule assessment. Adequate lead time for material and equipment procurements is essential. The evaluation of procurement advantages of using prefabricated, preassembled and modularized components is very important to be resolved early in order to avoid potential design impact. The source of the established durations for ordering, fabrication, and shipping time needs to be verified. “Just in time” delivery is an important concept to keep in mind during the planning. The impact of late deliveries is obvious in the delay potential for the project. The impact of “too early” delivery of materials and equipment can be just as significant: requiring additional storage and laydown areas; protection from the weather or damage in storage location; multiple handling of the material; etc. On more than one project, the constructor has incurred significant costs and inefficiencies related to longer than planned storage and warehousing.
- *Weather impact considerations.* An important element of the construction planning is to identify seasonal weather periods that could potentially impact the construction work. The potentially adverse weather time frames, when it is either too hot or too cold to work, should be included in the construction planning and incorporated into the schedule. Historic weather information is readily available and should be consulted when developing the construction schedule. For example the National Oceanic and Atmospheric Administration (NOAA) web site maintains a historic weather database for all regions of the United States. Consider using the scheduling software calendar features to factor in the potential adverse weather seasons. Construction schedule work activities that are susceptible to these weather impacts, particularly critical path activities should be assigned to that weather calendar. The completion of “facility enclosure” should occur prior to the inclement weather season in order to allow interior work to proceed without unnecessary delay or damage to the interior work. It is important to consider both maximum and minimum temperature requirements for material installation. Improper planning can result in costly extras for construction. Proper sequencing for the completion of climate control systems to provide temporary (or permanent) heating/cooling for the installation of climate sensitive materials such as sheetrock, millwork, and flooring.

- *Appropriateness of detail.* The construction schedule should not be a duplication of the project submittal register, expediting log, or the project completion punch list. Those are separate project management documents.

Site Planning Development

Site access and work restrictions need to be an element of the SCR. Work restrictions includes items such as allowable hours for construction noises; seasonal periods of non-work; labor or contract restrictions related to work on weekends or holidays; etc. There are several other elements related to the development of the site plan and specific site conditions that should be considered in the project planning. Specialists involved in the project need to be included in the planning for cranes; fire protection; safety; material management (procurement, transportation, delivery, and storage); and others. The site plan must include required spaces for the temporary facilities and material/equipment laydown areas and existing plant structures and features. The laydown areas should be located to allow the most direct access route possible to the final equipment locations and avoid multiple turns along the route. The site arrangement plan must be finalized early and should not be changed without valid reason. The following are important factors to consider in any schedule constructability assessment:^[2]

- *Temporary construction.* Establish a list and locations for the placement of required temporary facilities including: offices, warehouse(s), laydown areas, parking, tool cribs, fabrication shops (plan to locate adjacent to storage areas), medical facility & staff, welders' testing facility, chemical product (MSDS) storage, hazardous waste and combustible storage, etc. Layout the roads, access ways, fabrication shops and storage to avoid the cross flow of materials and minimize traffic interference. Consider outfitting "prepackaged" fabrication shops in large storage containers or trailers. These shops completely loaded with tools, equipment, and consumable supplies and could be readily moved to the work location and set on concrete work slabs. When practical, position the temporary facilities to take advantage of natural drainage patterns. The planning should also include removal of the temporary facilities at, or near the end of, the project construction phase to allow for any remaining construction work (grading, landscaping, paving, curbs, etc.) that may be required.
- *Site access planning.* Site access planning can result in significant savings of both labor cost and time. Location of the main gate should be centered on the construction work to minimize lost time due to travel. Depending on labor contract agreements (union and non-union) the project may be required to have separate craft entry gates and parking areas. Maximize open spaces for work areas, turnarounds, and maintenance. The width of roads and gates must be sized to accommodate the largest cranes and equipment that will be used or placed on the project. Staging areas should be established for items that are to be preassembled at the site before erection. These ground level preassembly construction activities increase productivity and enhance safety. Consider the use of man lifts in lieu of scaffolding where practical as scaffolding systems can be time consuming and expensive to erect and maintain. A daily review of site access & egress is required as the site can change daily as the installation work progresses.
- *Site plan.* The site arrangement plan should be reviewed along with the scheduled construction sequencing to be sure that the planned haul routes for heavy equipment are not closed because of height, width, length, and weight previously installed facilities/features or other restrictions. There should be a contingency plan for late delivery of large equipment so that other work can proceed. Review the site plan for equipment and material deliveries to avoid double handling by setting the items directly into position when received. As previously mentioned preassembly and modularization should be incorporated into the construction schedule as much as practical for the project.
- *Staffing and manpower planning.* A review of the manpower loading and craft staffing levels is an important element of the SCR. Consider the field craft manpower requirements in relation to the schedule and prioritize the engineering sequence of design releases to improve leveling of critical labor peaks. The required craft skills for the various phases of the project need to be compared with

the local craft labor availability during the planned construction phase. Other projects in the area that might compete for craft resources also need to be considered.

- *Optimum sequencing.* The construction sequence should be optimized so that there are not any engineering or procurement constraints. In other words, the project schedule should be construction driven and the engineering and procurement activities should support that construction schedule. Forecasted procurement durations for material/equipment deliveries must be evaluated and adjusted to support the construction timing and sequence requirements. Work that is not restricted by engineering or procurement lead-time should be pushed ahead quickly by construction.
- *Site safety.* Site safety planning includes providing clear access routes to prevent congestion as well as good escape or evacuation routes in case of fire or other emergencies on site. Minimize “hot work” permit areas and establish construction buffer areas away from existing operating facilities to reduce potential accidents involving facility operations personnel who normally should not be in the construction work areas.
- *Civil work sequencing.* The construction schedule should maximize the completion of underground work as early as practical on the project. Storm drains and sewage drain systems should be completed early. Temporary ditching and standby pumping units to remove surface water quickly should be included in the construction planning when appropriate. If possible the permanent plant sewage treatment facility should be completed early and placed into operation to minimize the need for temporary facilities. Drainage during construction excavation should be directed away from equipment foundations. It may be beneficial to complete parking areas early to reduce dust control requirements and provide parking for trades or additional storage for construction materials or equipment.
- *Permanent systems used during construction.* If permitted by the contract specifications, the permanent plant lighting should be completed as early as practical so that it can be used for night work, shift work, and security. Use as much of the permanent systems equipment (transformers, lighting panels, yard lighting) as possible for temporary power requirements. Temporary power should be installed underground to reduce exposure to construction equipment damages. Layout overhead temporary power lines along fence lines to minimize interference with crane booms, and other construction rigging. Covered storage should be provided with a controlled climate for sensitive electrical and instrumentation components. Permanent mechanical and piping systems should be used as much as possible for temporary services (compressed air, steam, water, etc.). In summary, maximize the use of permanent systems whenever possible or practical. Issues regarding warranties and vendor support must be taken into consideration and resolved before making the final decision to use permanent rather than temporary facility systems.

Schedule Constructability Review Checklist

This SCR checklist is intended to recommend a process and sequence to analyze and assess the construction planning to identify potential coordination issues, out of sequence work, missed work details, unrealistic activity durations, potential time delays, and inter-contractor coordination items. The SCR should include the construction site and work flow assessment as well as an analysis of the “work schedule” related to:

- abutting contractors
- limited work hours, restrictions on some construction activities such as blasting, trucking
- maintenance of traffic, restrictions on traffic flow and access to site
- disruption to public and businesses
- coordination with utilities and other agencies
- coordination with suppliers (long lead orders)
- subcontractors’ delays; contractors (ability) to effectively manage subs

- weather effects on schedule
- cash flow and contractor payments.”^[21]

Site Construction Schedule Checklist

A SCR checklist should generally follow the established sequence of construction. A constructability guide prepared by Edward D. Wright with Obrien-Kreitzberg Associates Inc. is very good resource that is available on the internet^[24]. Portions of that guide as related to schedule assessment have been included in the following proposed checklist:

Site Access

Getting into the site work area:

- construction permits
- environmental permits
- site clearing & grading requirements: location of disposal sites for spoils
- main entrance gate centrally located
- accessibility for construction personnel, equipment, and material delivery
- one way traffic pattern and parking for craft trades
- fences, barricades, traffic control, road closures, rerouting of normal traffic
- separate routes for construction traffic, vendor deliveries and construction vehicles
- utilities: relocation, installation of temporary utilities

Getting around the site interior work areas:

- accessibility for heavy equipment (crawler canes, etc.)
- facilities for construction field offices, client, subcontractors, and medical
- temporary power & utilities
- weather effects on work areas
- sewage & storm drainage system
- warehouse(s) & laydown areas
- fabrication shops
- modular equipment & components
- construction vehicle parking, maintenance and service area
- sign and markers

Site Civil Work

- clearing & grubbing
- site excavation & fill (ditches & trenches)
- storm drainage & sewers
- duct banks and underground utilities
- pile driving, shoring, retaining walls
- grading & paving

Facility Work

- foundations & footings
- concrete: cast in-place and pre-cast
- structural framework; windows; exterior siding, cladding, facades; & roof systems
- building enclosure
- mechanical & electrical rough-in

- interior walls & ceiling
- doors & hardware
- interior finishes; mill work, painting
- mechanical systems installation, cleaning, pressure testing, & final connection
- electrical conduit & cable tray loading, pulling cables, wiring connections & testing circuits
- furniture & fixtures
- system(s) checkout & startup: (i.e. HVAC system balancing, etc.)

REFERENCES

1. Bechtel Corporation, *The Project Schedule - Part 1 & 2*, Prime Contracts Management Bulletin No. 10, June 1991 & No. 11, December 1991, San Francisco, CA.
2. Bechtel Corporation, *Bechtel Constructability Handbook*, Rev. 0, 1992, San Francisco CA.
3. *Constructability Catalog*, Construction Engineering Research Laboratory, Champaign IL, www.cacer.army.mil
4. Construction Industry Institute (CII). *Constructability: A Primer*, Publication RS3-1 (July), CII, Austin, Texas, 1986.
5. Construction Industry Institute (CII). *Benefits and Costs of Constructability*, CII Publication SD-83-1992.
6. Construction Management Association of America (CMAA). *Fifth Annual Survey of Owners Presentation September 2004*, World-wide Web document, http://cmaanet.org/fmi_survey.php, (January 3, 2005).
7. Dovichi, Stephanie L., *A Lesson in Constructability 101*, Coalition for Adequate School Housing (CASH), 2002, Sacramento, CA, www.cashnet.org June 2004.
8. Folk, P. Douglas, Esq. *Constructability: Does the Design Convey Enough Information?*, Schinner 44th Annual Meeting of Invited Attorneys, 2005.
9. Gransberg, Douglas D., *Implementing Project Constructability*, AACE International Transactions, AACE International, Morgantown, WV, 2005.
10. Gibson, G.E., C.I. McGinnis, W.S. Flanigan, and J.E. Wood, *Constructability in the Public Sector*, J. of Constr. Engr. and Mngt. ASCE, 122 (3), New York, NY: 274-280, 1996.
11. Hanlon, Eric J., *Constructability Information Classification Scheme*, American Society of Civil Engineers (ASCE), 1995.
12. *Heat Recovery Steam Generator – HRSGs – Constructability*, www.babcockpower.com, April 14, 2007
13. Jergeas, G. and J. Van der Put, *Benefits of Constructability on Construction Projects*, Journal of Construction Engineering and Management, ASCE, Vol. 127, (4), 2001.
14. Keville Enterprises, *Constructability Reviews*, www.keville.com/constructability_reviews, June 2006.
15. KJM & Associates, *Constructability Reviews*, www.kjmassoc.com, June 2004.
16. McDonald, Donald F. Jr., *Weather Delays and Impacts*, Cost Engineering journal, AACE International, Morgantown, WV, May 2000.
17. Pocock, James B., *Constructability State of Practice*, Means, Methods & Trends, AEI/CI, 2005, www.mmtmagazine.org, April, 17, 2007
18. Ruby, David I, Ruby, Jay, & Volpe, Brian M.; *Constructability: A Design Philosophy to Build On*, Modern Steel Construction, April 2007.
19. Staub-French, Sheryl, *Providing Cost and Constructability Feedback to Designers*, Construction Research Congress 2003.
20. Thabet, Walid, *Design/Construction Integration thru Virtual Construction for Improved Constructability*, White Paper, Berkley-Stanford CEM Workshop, August 1999, Stanford CA.
21. Touran, Ali, CMAA, *Owners Risk Reduction Techniques using a CM, Checklist for Capital Construction Projects*, Northeastern University. October 2006
22. U.S. Army Corps of Engineers (USACE), *Engineer Regulation 415-1-11: Biddability, Constructability, Operability, and Environmental Review*, Department of the Army, Washington, D.C., 1994.
23. Vanir Construction Management Services, *Constructability Reviews*, www.vanir.com/services/ConstructabilityReview, June 2006

24. Wright, Edward D. P.E., Obrien-Kreitzberg Assoc. Inc, *Constructability Guide*, March 1994, www.azdot.gov/highways/constgrp/PDF
25. AACE International Recommended Practice No. 10S-90 *Cost Engineering Terminology*, AACE International, Morgantown, WV, (latest revision).
26. AACE International Recommended Practice No. 30R-03 *Implementing Project Constructability*, AACE International, Morgantown, WV, (latest revision).
27. Hollmann, John K., Editor, *Total Cost Management Framework: An Integrated Approach to Portfolio, Program, and Project Management*, AACE International, Morgantown WV, 2006.
28. AIA Integrated Project Delivery: A Guide, American Institute of Architects Contract Documents Committee, http://info.aia.org/SiteObjects/files/IPD_Guide_2007.pdf
29. McGraw-Hill Construction, *Best Practices in Integrated Project Delivery for Overall Improved Service Delivery Management*, [www.http://ce.construction.com](http://www.ce.construction.com), June 2009
30. Integrated Project Delivery (IPD), <http://www.kalblue.com/construction/bim/integrated-project-delivery-ipd>, June 2009

CONTRIBUTORS

Edward E. Douglas, III CCC PSP (Author)
Ricardo Accioly
Rodney B. Adams, CCE
Zia Akhtar
Kenneth Baker
Michael P. Bomi
Timothy T. Calvey, PE PSP
Paul L. Conant, Jr.
Douglas A. Findley, CCC PSP
Ricardo Garcia da Roza
John K. Hollmann, PE CCE CEP
John J. MacDonald
Robert C. Powell, CCE
Benjamin Price
Mahmoud M. Saleh
Hannah E. Schumacher, PSP
Donald F. Sulzer
Ronald M. Winter, PSP